
$f = x^2 - 2$, the number of post critical points is 2

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: $x[1]$

-> Checking Foot $x[2]$: Attached poles in G: ['p[0]']

-> Pathfinding $x[1]$ to $x[2]$: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[1]$

Leg to $x[2]$: $x[1] \rightarrow d[0] \rightarrow x[2]$

Evaluating Head Candidate: $x[2]$

-> Checking Foot $x[1]$: Attached poles in G: ['p[1]']

-> Pathfinding $x[2]$ to $x[1]$: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[2]$

Leg to $x[1]$: $x[2] \rightarrow d[0] \rightarrow x[1]$

$f = x^2 - 1$, the number of post critical points is 2

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: $x[1]$

-> Checking Foot $x[2]$: Attached poles in G: ['p[0]']

-> Pathfinding $x[1]$ to $x[2]$: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[1]$

Leg to $x[2]$: $x[1] \rightarrow p[0] \rightarrow x[2]$

Evaluating Head Candidate: $x[2]$

-> Checking Foot $x[1]$: Attached poles in G: ['p[1]', 'p[0]']

[REJECTED] Foot $x[1]$ has > 1 pole. Head $x[2]$ is invalid.

$f = 1/2/(x - 1)^2$, the number of post critical points is 4

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: $x[1]$

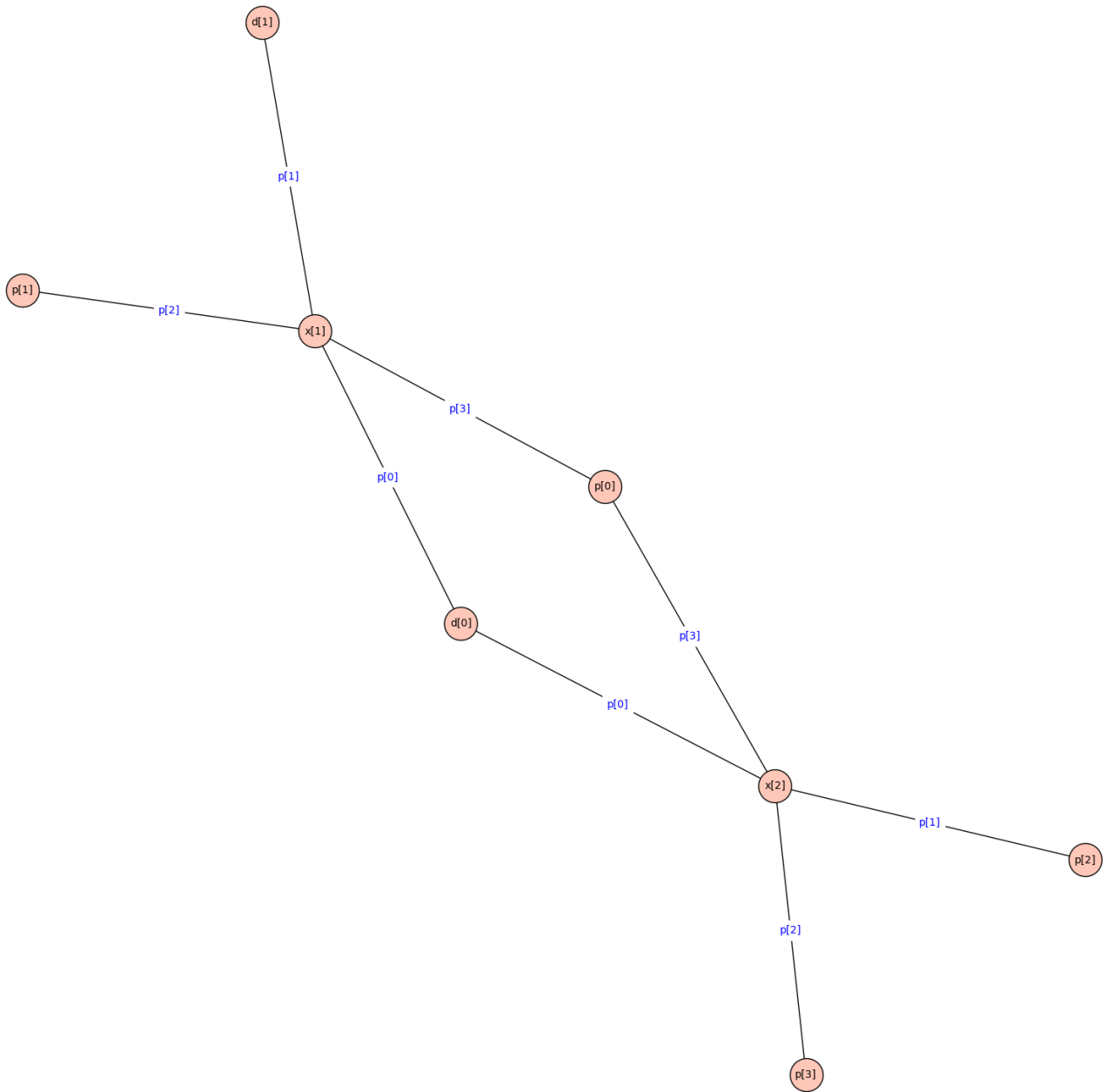
-> Checking Foot $x[2]$: Attached poles in G: ['p[2]', 'p[3]', 'p[0]']

[REJECTED] Foot $x[2]$ has > 1 pole. Head $x[1]$ is invalid.

Evaluating Head Candidate: $x[2]$

-> Checking Foot x[1]: Attached poles in G: ['p[1]', 'p[0]']
 [REJECTED] Foot x[1] has > 1 pole. Head x[2] is invalid.

No valid spider found linking a head to all feet under the strict pole rules.
 Displaying the graph...



$f = (x - 1)^{-2}$, the number of post critical points is 4

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: ['p[2]', 'p[3]', 'p[0]']
 [REJECTED] Foot x[2] has > 1 pole. Head x[1] is invalid.

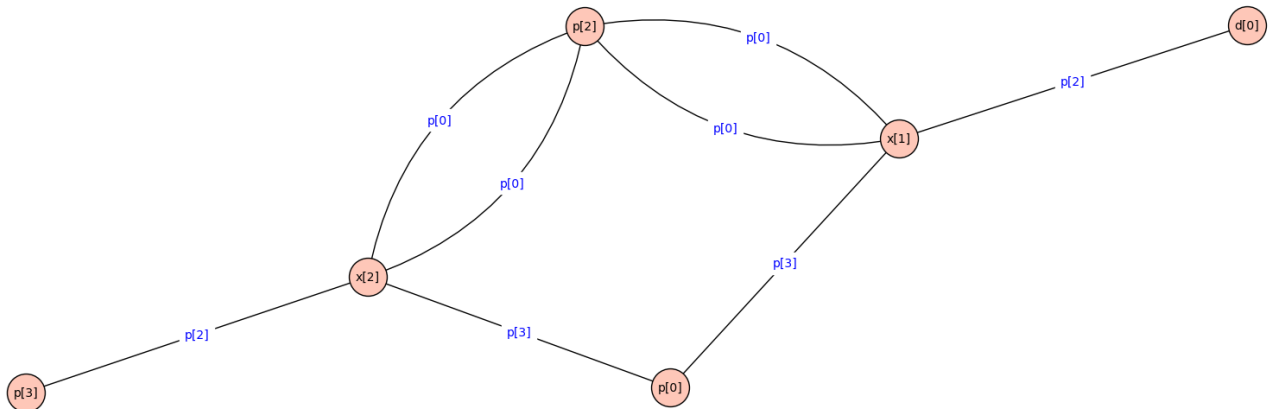
Evaluating Head Candidate: $x[2]$

→ Checking Foot $x[1]$: Attached poles in G: ['p[2]', 'p[0]']

[REJECTED] Foot $x[1]$ has > 1 pole. Head $x[2]$ is invalid.

No valid spider found linking a head to all feet under the strict pole rules.

Displaying the graph...



$f = -1/4/(x^2 - x)$, the number of post critical points is 4

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: $x[1]$

→ Checking Foot $x[2]$: Attached poles in G: ['p[2]', 'p[1]']

[REJECTED] Foot $x[2]$ has > 1 pole. Head $x[1]$ is invalid.

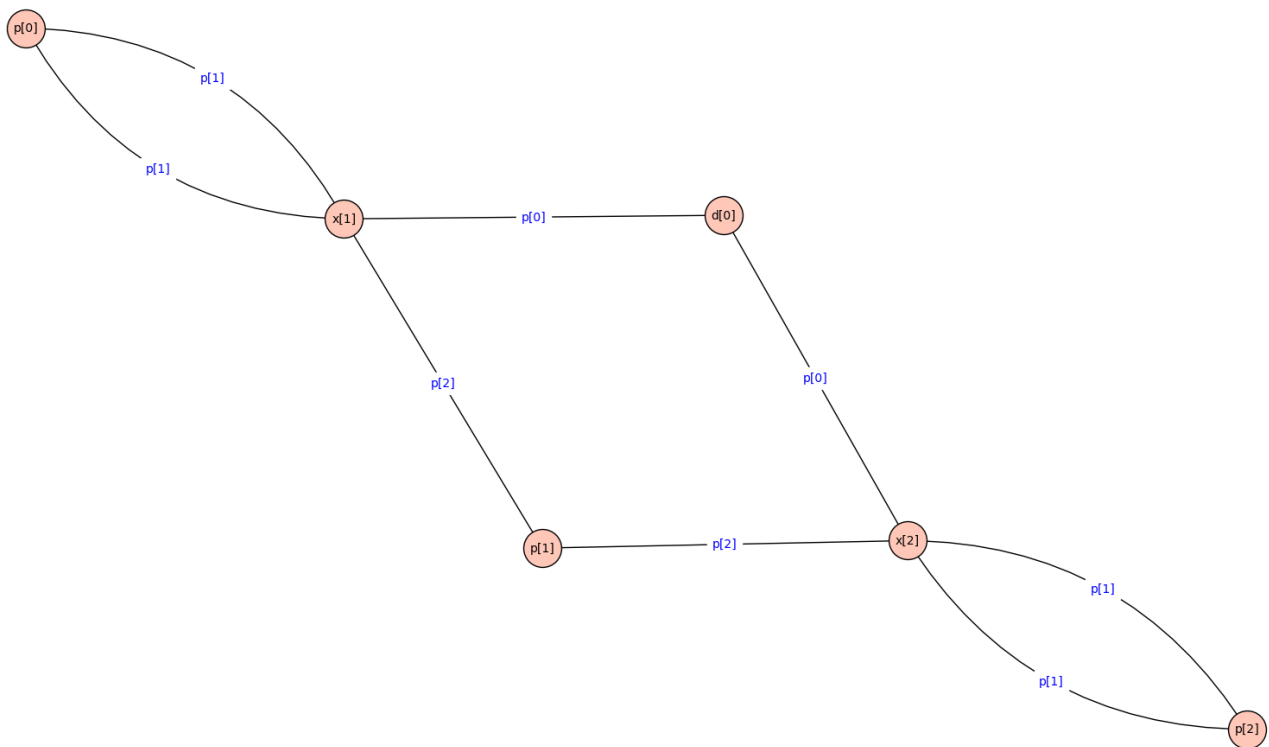
Evaluating Head Candidate: $x[2]$

→ Checking Foot $x[1]$: Attached poles in G: ['p[0]', 'p[1]']

[REJECTED] Foot $x[1]$ has > 1 pole. Head $x[2]$ is invalid.

No valid spider found linking a head to all feet under the strict pole rules.

Displaying the graph...



$f = -4/3/(3*x^2 - 4*x)$, the number of post critical points is 5

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: x[1]

→ Checking Foot x[2]: Attached poles in G: ['p[2]', 'p[0]']

[REJECTED] Foot x[2] has > 1 pole. Head x[1] is invalid.

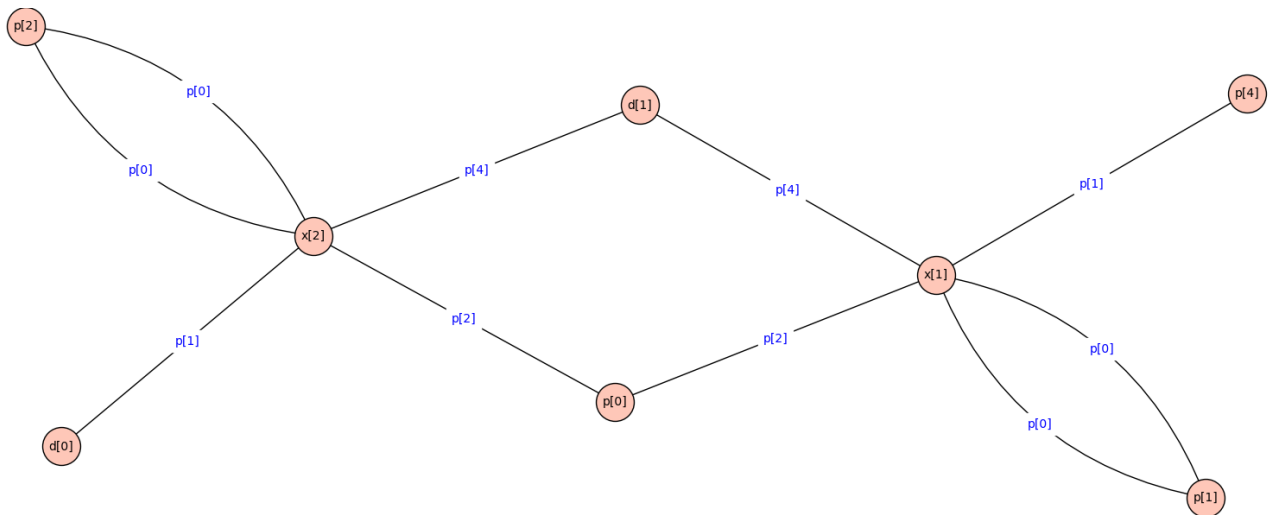
Evaluating Head Candidate: x[2]

→ Checking Foot x[1]: Attached poles in G: ['p[1]', 'p[4]', 'p[0]']

[REJECTED] Foot x[1] has > 1 pole. Head x[2] is invalid.

No valid spider found linking a head to all feet under the strict pole rules.

Displaying the graph...



$f = 2/(x - 1)^2$, the number of post critical points is 3

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: $x[1]$

→ Checking Foot $x[2]$: Attached poles in G: ['p[2]', 'p[0]']

[REJECTED] Foot $x[2]$ has > 1 pole. Head $x[1]$ is invalid.

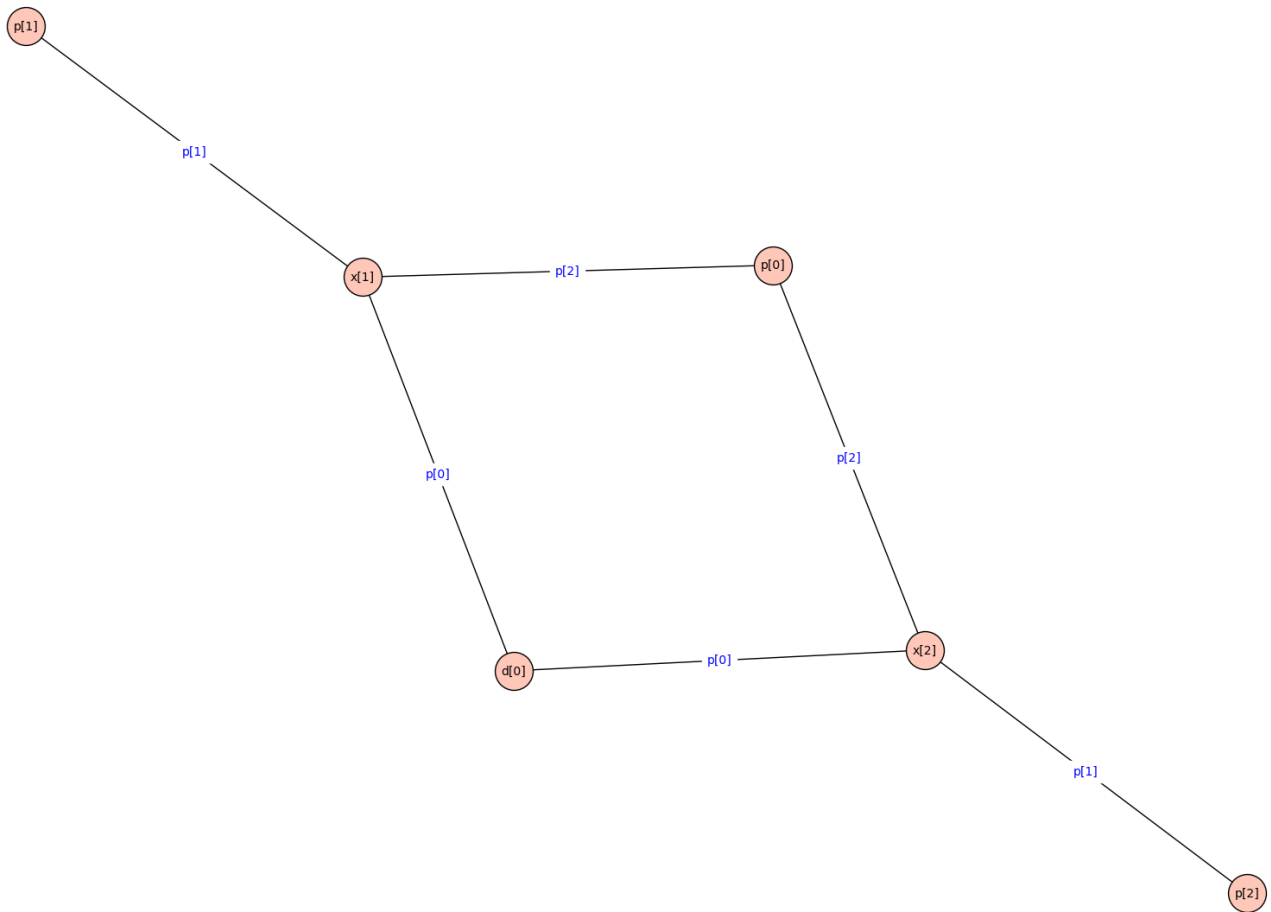
Evaluating Head Candidate: $x[2]$

→ Checking Foot $x[1]$: Attached poles in G: ['p[1]', 'p[0]']

[REJECTED] Foot $x[1]$ has > 1 pole. Head $x[2]$ is invalid.

No valid spider found linking a head to all feet under the strict pole rules.

Displaying the graph...



$f = -1/2*(2*x + 1)/(x^2 - 2*x)$, the number of post critical points is 6

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: $x[1]$

-> Checking Foot $x[2]$: Attached poles in G: ['p[1]']

-> Pathfinding $x[1]$ to $x[2]$: Found 3 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[1]$

Leg to $x[2]$: $x[1] \rightarrow d[0] \rightarrow x[2]$

Evaluating Head Candidate: $x[2]$

-> Checking Foot $x[1]$: Attached poles in G: ['p[4]']

-> Pathfinding $x[2]$ to $x[1]$: Found 3 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[2]$

Leg to $x[1]$: $x[2] \rightarrow d[0] \rightarrow x[1]$

$f = -2*x/(2*x^2 - 4*x + 1)$, the number of post critical points is 4

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: ['p[3]']

-> Pathfinding x[1] to x[2]: Found 3 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

Leg to x[2]: x[1] -> d[2] -> x[2]

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: ['p[3]']

-> Pathfinding x[2] to x[1]: Found 3 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

Leg to x[1]: x[2] -> d[2] -> x[1]

$f = -(3x^2 - 4x + 1)/(4x - 1)$, the number of post critical points is 5

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: ['p[0]', 'p[4]', 'p[2]']

[REJECTED] Foot x[2] has > 1 pole. Head x[1] is invalid.

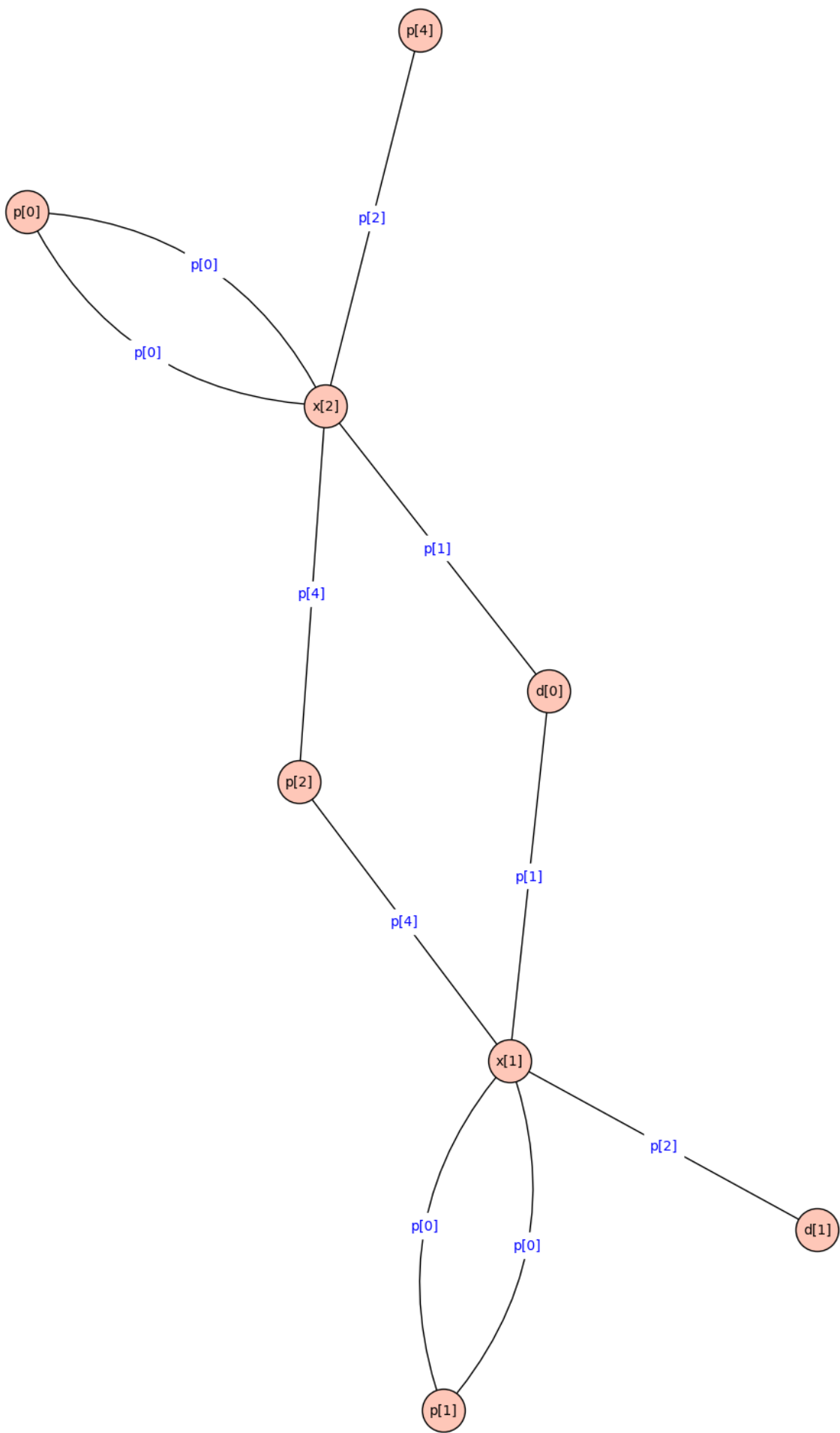
Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: ['p[1]', 'p[2]']

[REJECTED] Foot x[1] has > 1 pole. Head x[2] is invalid.

No valid spider found linking a head to all feet under the strict pole rules.

Displaying the graph...



$f = 1/2 * x + 1/x$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: $x[1]$

-> Checking Foot $x[2]$: Attached poles in G: []

-> Pathfinding $x[1]$ to $x[2]$: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[1]$

Leg to $x[2]$: $x[1] \rightarrow d[0] \rightarrow x[2]$

Evaluating Head Candidate: $x[2]$

-> Checking Foot $x[1]$: Attached poles in G: []

-> Pathfinding $x[2]$ to $x[1]$: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[2]$

Leg to $x[1]$: $x[2] \rightarrow d[0] \rightarrow x[1]$

$f = 1/2 * x + 1/2/x$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: $x[1]$

-> Checking Foot $x[2]$: Attached poles in G: ['p[0]', 'p[1]']

[REJECTED] Foot $x[2]$ has > 1 pole. Head $x[1]$ is invalid.

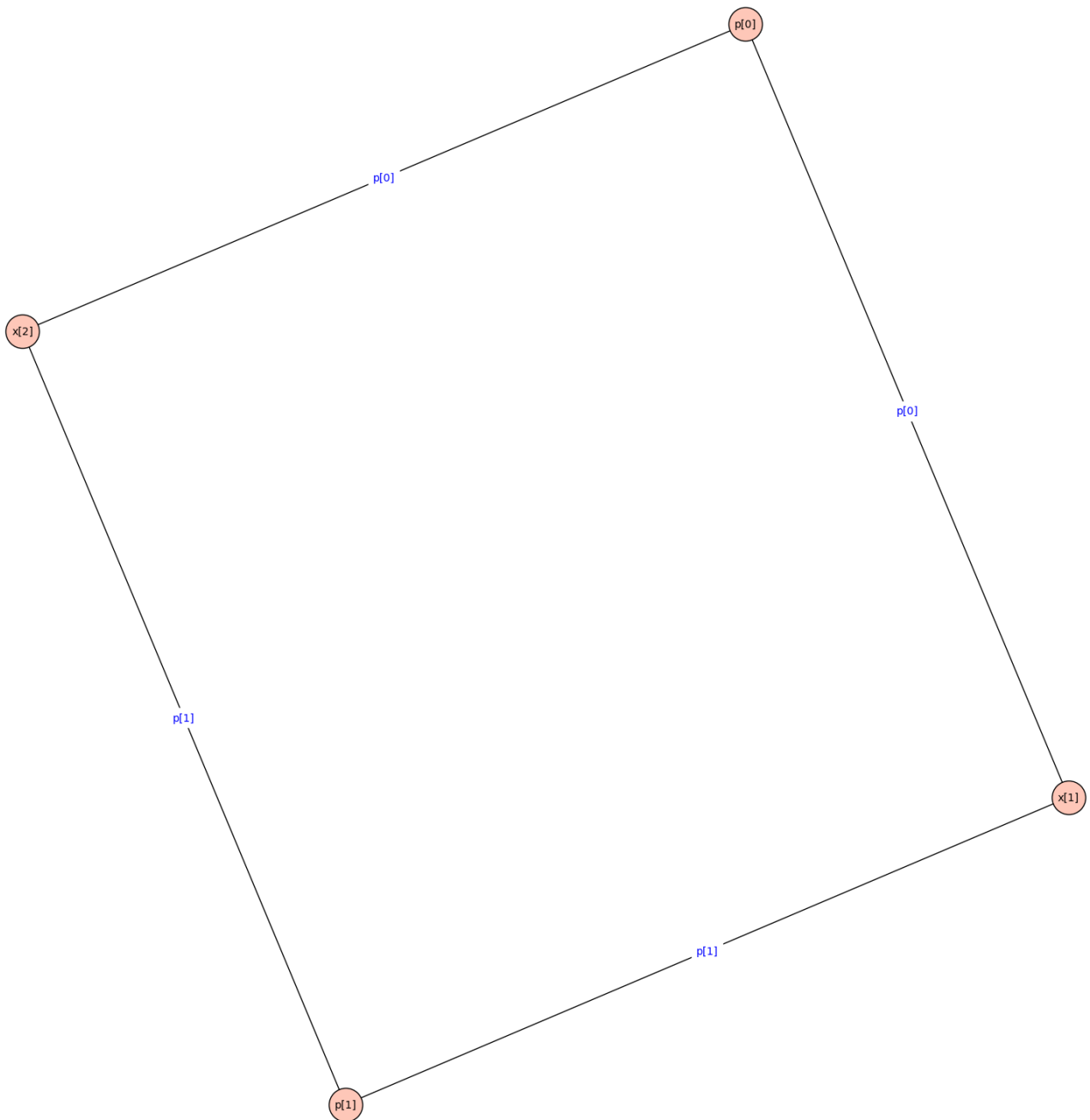
Evaluating Head Candidate: $x[2]$

-> Checking Foot $x[1]$: Attached poles in G: ['p[0]', 'p[1]']

[REJECTED] Foot $x[1]$ has > 1 pole. Head $x[2]$ is invalid.

No valid spider found linking a head to all feet under the strict pole rules.

Displaying the graph...



$f = 1/2 * x - 3/2 / x$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: []

-> Pathfinding x[1] to x[2]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

Leg to x[2]: x[1] -> d[0] -> x[2]

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: []
-> Pathfinding x[2] to x[1]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]
Leg to x[1]: x[2] -> d[0] -> x[1]

$f = 1/2 * x - 1/2/x$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]
-> Checking Foot x[2]: Attached poles in G: []
-> Pathfinding x[1] to x[2]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]
Leg to x[2]: x[1] -> d[0] -> x[2]

Evaluating Head Candidate: x[2]
-> Checking Foot x[1]: Attached poles in G: []
-> Pathfinding x[2] to x[1]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]
Leg to x[1]: x[2] -> d[0] -> x[1]

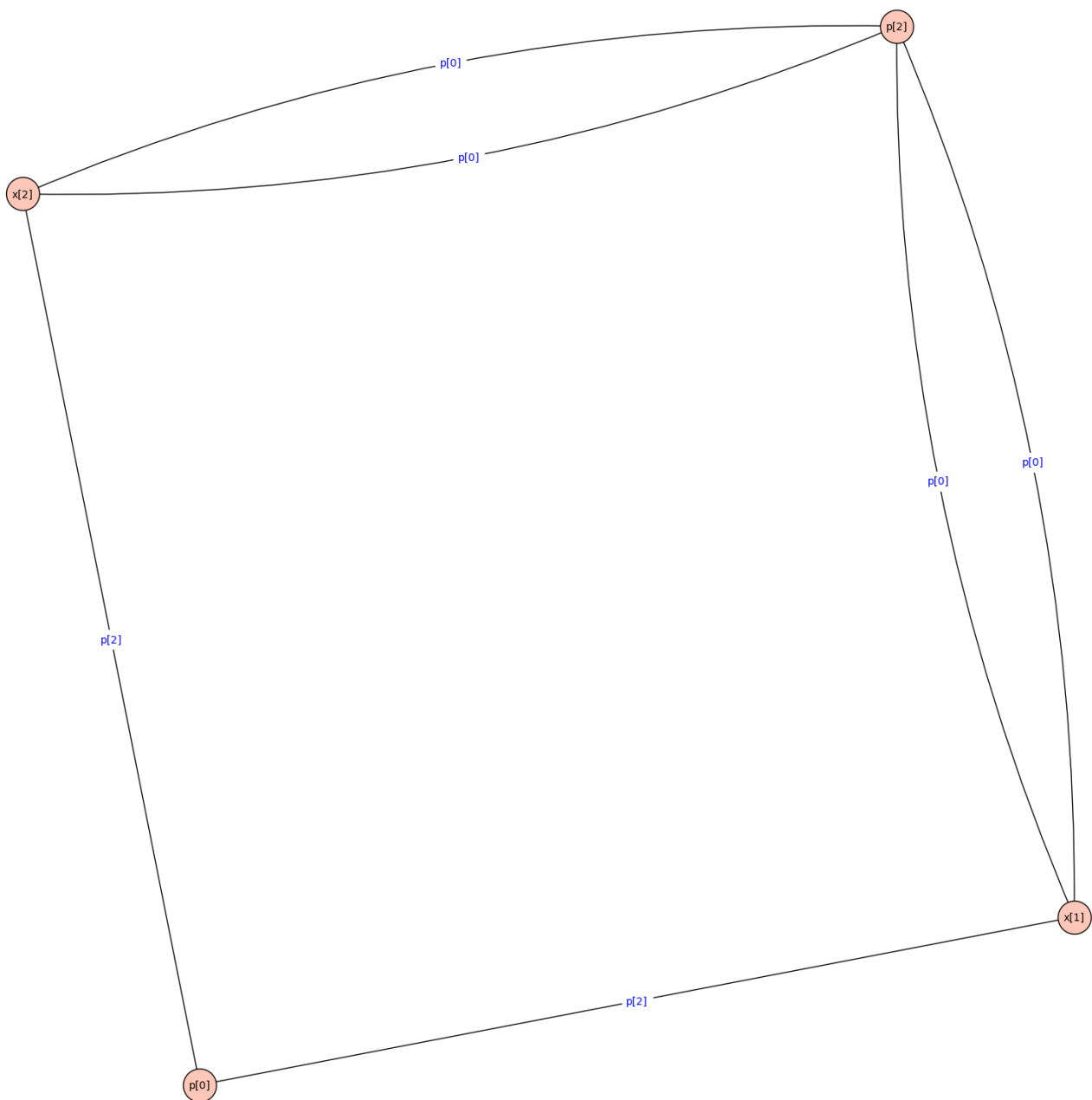
$f = x^{(-2)}$, the number of post critical points is 3

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]
-> Checking Foot x[2]: Attached poles in G: ['p[2]', 'p[0]']
[REJECTED] Foot x[2] has > 1 pole. Head x[1] is invalid.

Evaluating Head Candidate: x[2]
-> Checking Foot x[1]: Attached poles in G: ['p[2]', 'p[0]']
[REJECTED] Foot x[1] has > 1 pole. Head x[2] is invalid.

No valid spider found linking a head to all feet under the strict pole rules.
Displaying the graph...



$f = 2/x^2$, the number of post critical points is 3

---- Spider Search Results (DIAGNOSTIC MODE) ----

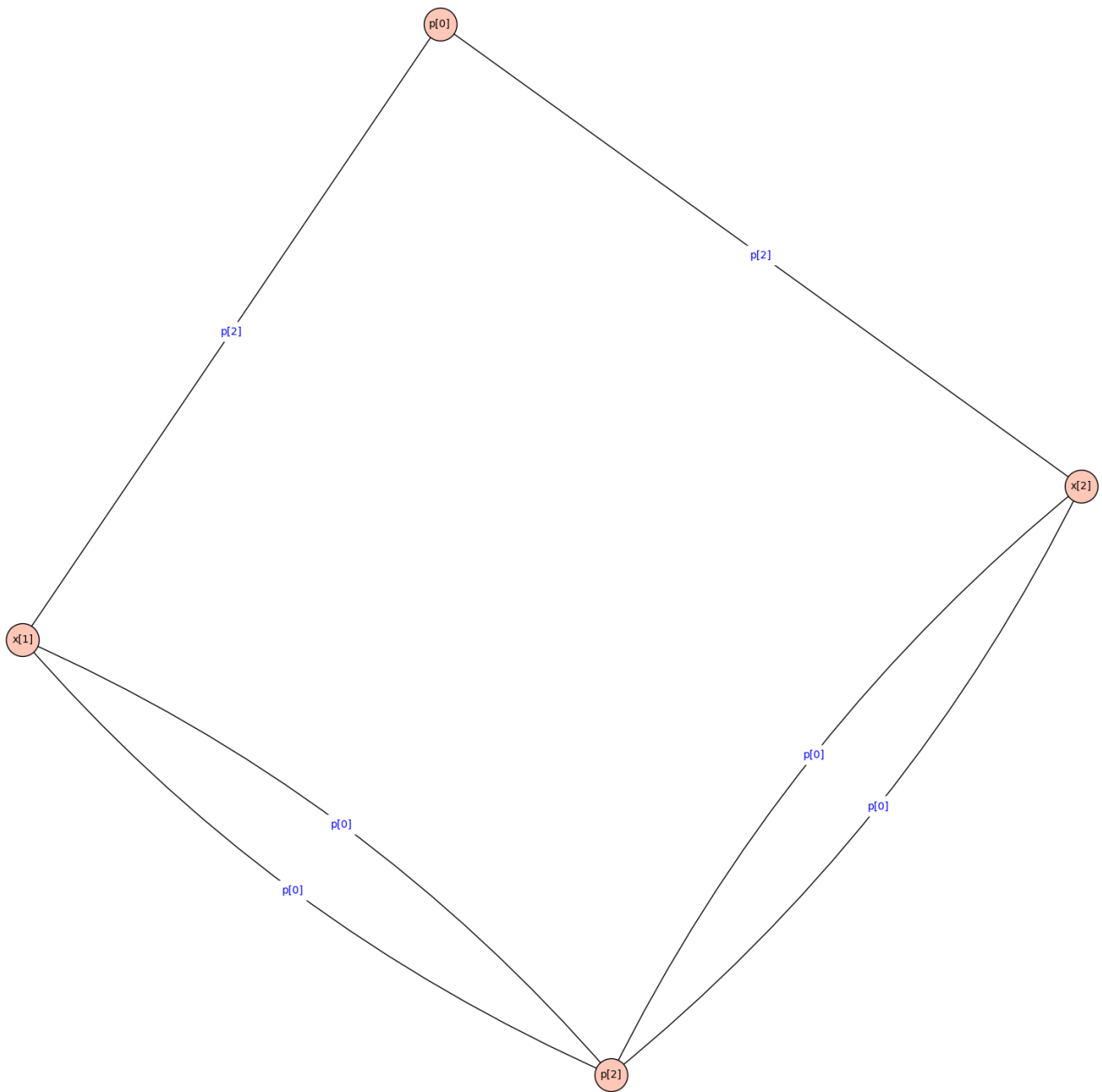
Evaluating Head Candidate: $x[1]$

→ Checking Foot $x[2]$: Attached poles in G: ['p[2]', 'p[0]']
 [REJECTED] Foot $x[2]$ has > 1 pole. Head $x[1]$ is invalid.

Evaluating Head Candidate: $x[2]$

→ Checking Foot $x[1]$: Attached poles in G: ['p[2]', 'p[0]']
 [REJECTED] Foot $x[1]$ has > 1 pole. Head $x[2]$ is invalid.

No valid spider found linking a head to all feet under the strict pole rules.
Displaying the graph...



$f = -(x^2 - 2x - 1)/(x^2 + 2x - 1)$, the number of post critical points is 2

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: []

-> Pathfinding x[1] to x[2]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

Leg to x[2]: x[1] -> d[0] -> x[2]

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: []

-> Pathfinding x[2] to x[1]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

Leg to x[1]: x[2] -> d[0] -> x[1]

$f = -(x - 2)*x/(2*x - 1)$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: []

-> Pathfinding x[1] to x[2]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

Leg to x[2]: x[1] -> d[0] -> x[2]

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: []

-> Pathfinding x[2] to x[1]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

Leg to x[1]: x[2] -> d[0] -> x[1]

$f = (2*x - 1)/(x^2 - 1)$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: []

-> Pathfinding x[1] to x[2]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

Leg to x[2]: x[1] -> d[0] -> x[2]

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: []

-> Pathfinding x[2] to x[1]: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

Leg to $x[1]$: $x[2] \rightarrow d[0] \rightarrow x[1]$

$f = (x^2 - 4x + 2)/(x^2 - 2x + 2)$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: $x[1]$

-> Checking Foot $x[2]$: Attached poles in G: []

-> Pathfinding $x[1]$ to $x[2]$: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[1]$

Leg to $x[2]$: $x[1] \rightarrow d[0] \rightarrow x[2]$

Evaluating Head Candidate: $x[2]$

-> Checking Foot $x[1]$: Attached poles in G: []

-> Pathfinding $x[2]$ to $x[1]$: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[2]$

Leg to $x[1]$: $x[2] \rightarrow d[0] \rightarrow x[1]$

$f = -4x/(x^2 + 2)$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: $x[1]$

-> Checking Foot $x[2]$: Attached poles in G: []

-> Pathfinding $x[1]$ to $x[2]$: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[1]$

Leg to $x[2]$: $x[1] \rightarrow d[0] \rightarrow x[2]$

Evaluating Head Candidate: $x[2]$

-> Checking Foot $x[1]$: Attached poles in G: []

-> Pathfinding $x[2]$ to $x[1]$: Found 2 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[2]$

Leg to $x[1]$: $x[2] \rightarrow d[0] \rightarrow x[1]$

$f = x^3$, the number of post critical points is 1

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

- > Checking Foot x[2]: Attached poles in G: ['p[0]']
- > Checking Foot x[3]: Attached poles in G: ['p[0]']
- > Pathfinding x[1] to x[2]: Found 1 simple paths.
- > Pathfinding x[1] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

- Leg to x[2]: x[1] -> p[0] -> x[2]
- Leg to x[3]: x[1] -> p[0] -> x[3]

Evaluating Head Candidate: x[2]

- > Checking Foot x[1]: Attached poles in G: ['p[0]']
- > Checking Foot x[3]: Attached poles in G: ['p[0]']
- > Pathfinding x[2] to x[1]: Found 1 simple paths.
- > Pathfinding x[2] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

- Leg to x[1]: x[2] -> p[0] -> x[1]
- Leg to x[3]: x[2] -> p[0] -> x[3]

Evaluating Head Candidate: x[3]

- > Checking Foot x[1]: Attached poles in G: ['p[0]']
- > Checking Foot x[2]: Attached poles in G: ['p[0]']
- > Pathfinding x[3] to x[1]: Found 1 simple paths.
- > Pathfinding x[3] to x[2]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[3]

- Leg to x[1]: x[3] -> p[0] -> x[1]
- Leg to x[2]: x[3] -> p[0] -> x[2]

$f = -x^3 + 1$, the number of post critical points is 2

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: x[1]

- > Checking Foot x[2]: Attached poles in G: ['p[0]']
- > Checking Foot x[3]: Attached poles in G: ['p[1]', 'p[0]']
- [REJECTED] Foot x[3] has > 1 pole. Head x[1] is invalid.

Evaluating Head Candidate: x[2]

- > Checking Foot x[1]: Attached poles in G: ['p[0]']
- > Checking Foot x[3]: Attached poles in G: ['p[1]', 'p[0]']
- [REJECTED] Foot x[3] has > 1 pole. Head x[2] is invalid.

Evaluating Head Candidate: x[3]

- > Checking Foot x[1]: Attached poles in G: ['p[0]']
- > Checking Foot x[2]: Attached poles in G: ['p[0]']
- > Pathfinding x[3] to x[1]: Found 1 simple paths.
- > Pathfinding x[3] to x[2]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[3]

- Leg to x[1]: x[3] -> p[0] -> x[1]
- Leg to x[2]: x[3] -> p[0] -> x[2]

$f = -2x^3 + 3x^2 + 1/2$, the number of post critical points is 3

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: x[1]

- > Checking Foot x[2]: Attached poles in G: ['p[1]', 'p[2]']
- [REJECTED] Foot x[2] has > 1 pole. Head x[1] is invalid.

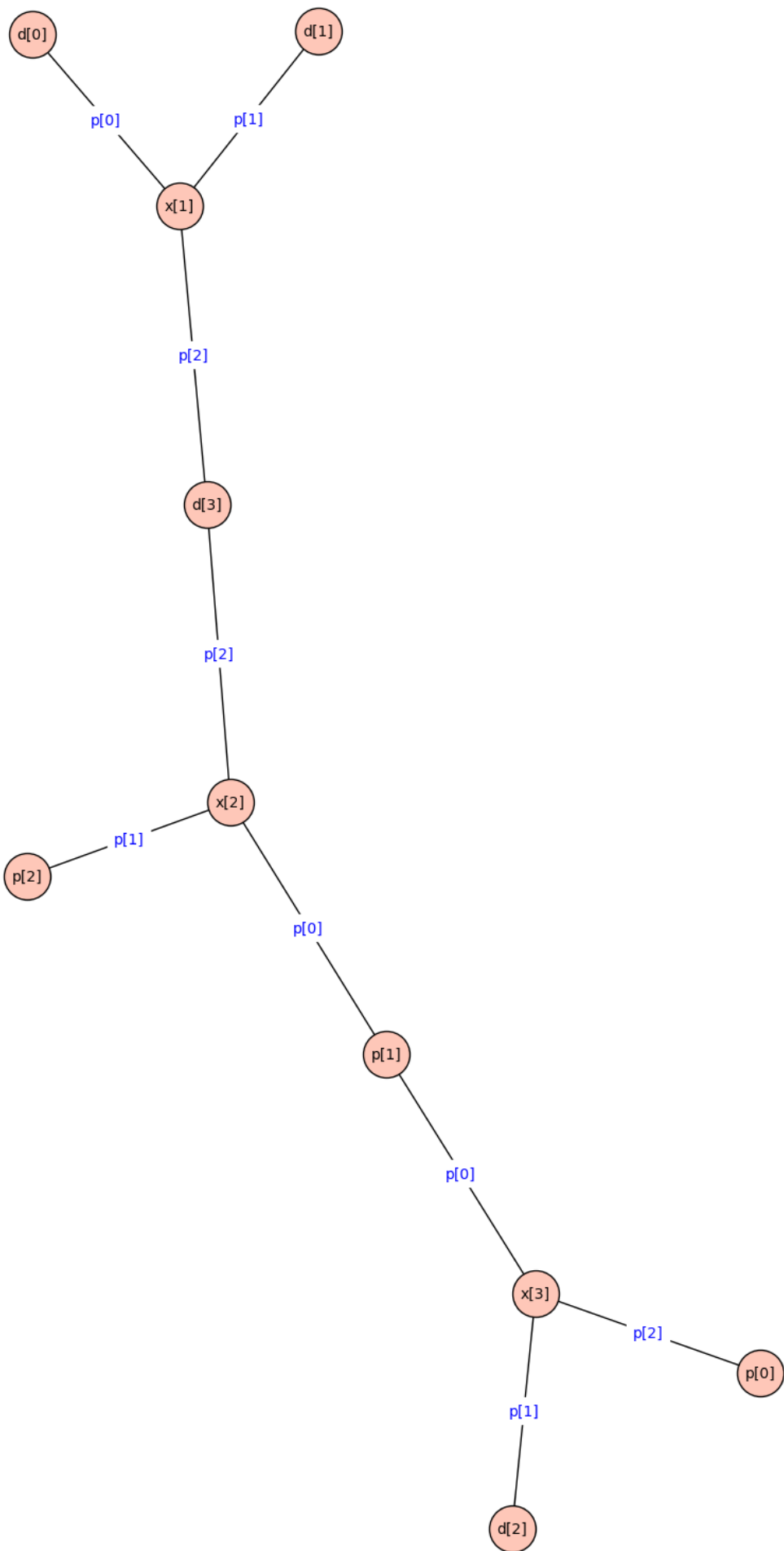
Evaluating Head Candidate: x[2]

- > Checking Foot x[1]: Attached poles in G: []
- > Checking Foot x[3]: Attached poles in G: ['p[1]', 'p[0]']
- [REJECTED] Foot x[3] has > 1 pole. Head x[2] is invalid.

Evaluating Head Candidate: x[3]

- > Checking Foot x[1]: Attached poles in G: []
- > Checking Foot x[2]: Attached poles in G: ['p[1]', 'p[2]']
- [REJECTED] Foot x[2] has > 1 pole. Head x[3] is invalid.

No valid spider found linking a head to all feet under the strict pole rules.
Displaying the graph...



$f = -2x^3 + 3x^2$, the number of post critical points is 2

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: $x[1]$

→ Checking Foot $x[2]$: Attached poles in G: ['p[0]', 'p[1]']
[REJECTED] Foot $x[2]$ has > 1 pole. Head $x[1]$ is invalid.

Evaluating Head Candidate: $x[2]$

→ Checking Foot $x[1]$: Attached poles in G: ['p[0]']
→ Checking Foot $x[3]$: Attached poles in G: ['p[1]']
→ Pathfinding $x[2]$ to $x[1]$: Found 1 simple paths.
→ Pathfinding $x[2]$ to $x[3]$: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[2]$

Leg to $x[1]$: $x[2] \rightarrow p[0] \rightarrow x[1]$

Leg to $x[3]$: $x[2] \rightarrow p[1] \rightarrow x[3]$

Evaluating Head Candidate: $x[3]$

→ Checking Foot $x[1]$: Attached poles in G: ['p[0]']
→ Checking Foot $x[2]$: Attached poles in G: ['p[0]', 'p[1]']
[REJECTED] Foot $x[2]$ has > 1 pole. Head $x[3]$ is invalid.

$f = -x^3 + 3/2x^2 - 1$, the number of post critical points is 3

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: $x[1]$

→ Checking Foot $x[2]$: Attached poles in G: ['p[2]', 'p[1]']
[REJECTED] Foot $x[2]$ has > 1 pole. Head $x[1]$ is invalid.

Evaluating Head Candidate: $x[2]$

→ Checking Foot $x[1]$: Attached poles in G: []
→ Checking Foot $x[3]$: Attached poles in G: ['p[0]']
→ Pathfinding $x[2]$ to $x[1]$: Found 1 simple paths.
→ Pathfinding $x[2]$ to $x[3]$: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[2]$

Leg to $x[1]$: $x[2] \rightarrow d[3] \rightarrow x[1]$

Leg to $x[3]$: $x[2] \rightarrow d[3] \rightarrow x[1] \rightarrow d[2] \rightarrow x[3]$

Evaluating Head Candidate: $x[3]$

→ Checking Foot $x[1]$: Attached poles in G: []

-> Checking Foot $x[2]$: Attached poles in G: ['p[2]', 'p[1]']
 [REJECTED] Foot $x[2]$ has > 1 pole. Head $x[3]$ is invalid.

$f = 2 \cdot x^3 - 3 \cdot x^2 + 1/2$, the number of post critical points is 3

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: $x[1]$

-> Checking Foot $x[2]$: Attached poles in G: ['p[1]', 'p[0]']
 [REJECTED] Foot $x[2]$ has > 1 pole. Head $x[1]$ is invalid.

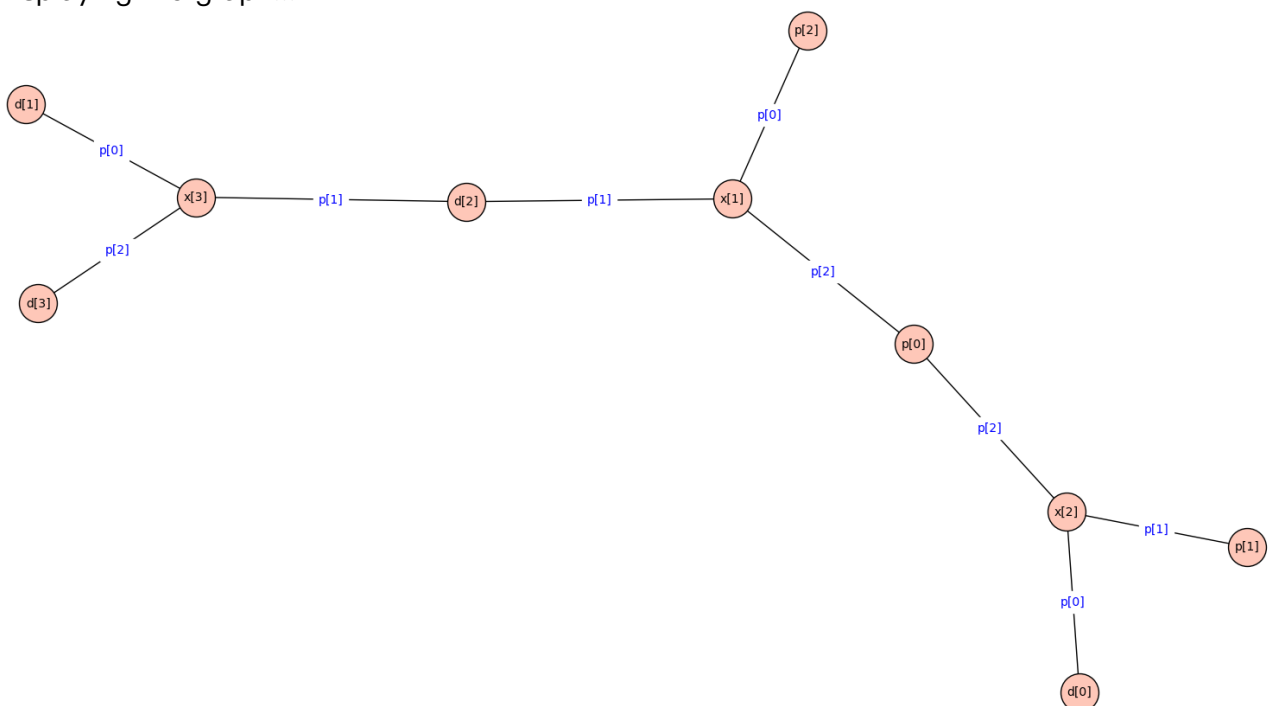
Evaluating Head Candidate: $x[2]$

-> Checking Foot $x[1]$: Attached poles in G: ['p[2]', 'p[0]']
 [REJECTED] Foot $x[1]$ has > 1 pole. Head $x[2]$ is invalid.

Evaluating Head Candidate: $x[3]$

-> Checking Foot $x[1]$: Attached poles in G: ['p[2]', 'p[0]']
 [REJECTED] Foot $x[1]$ has > 1 pole. Head $x[3]$ is invalid.

No valid spider found linking a head to all feet under the strict pole rules.
 Displaying the graph...



$f = x^3 - 3/2 \cdot x^2$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: ['p[0]', 'p[1]']
[REJECTED] Foot x[2] has > 1 pole. Head x[1] is invalid.

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: ['p[0]']
-> Checking Foot x[3]: Attached poles in G: []
-> Pathfinding x[2] to x[1]: Found 1 simple paths.
-> Pathfinding x[2] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

Leg to x[1]: x[2] -> p[0] -> x[1]
Leg to x[3]: x[2] -> p[0] -> x[1] -> d[1] -> x[3]

Evaluating Head Candidate: x[3]

-> Checking Foot x[1]: Attached poles in G: ['p[0]']
-> Checking Foot x[2]: Attached poles in G: ['p[0]', 'p[1]']
[REJECTED] Foot x[2] has > 1 pole. Head x[3] is invalid.

$f = -3x^3 + 9/2x^2$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: ['p[0]']
-> Checking Foot x[3]: Attached poles in G: ['p[1]']
-> Pathfinding x[1] to x[2]: Found 1 simple paths.
-> Pathfinding x[1] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

Leg to x[2]: x[1] -> p[0] -> x[2]
Leg to x[3]: x[1] -> p[0] -> x[2] -> d[1] -> x[3]

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: ['p[0]']
-> Checking Foot x[3]: Attached poles in G: ['p[1]']
-> Pathfinding x[2] to x[1]: Found 1 simple paths.
-> Pathfinding x[2] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

Leg to x[1]: x[2] -> p[0] -> x[1]
Leg to x[3]: x[2] -> d[1] -> x[3]

Evaluating Head Candidate: x[3]

-> Checking Foot x[1]: Attached poles in G: ['p[0]']
-> Checking Foot x[2]: Attached poles in G: ['p[0]']
-> Pathfinding x[3] to x[1]: Found 1 simple paths.
-> Pathfinding x[3] to x[2]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[3]
Leg to x[1]: x[3] -> d[1] -> x[2] -> p[0] -> x[1]
Leg to x[2]: x[3] -> d[1] -> x[2]

$f = -4x^3 + 6x^2 - 1/2$, the number of post critical points is 2

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]
-> Checking Foot x[2]: Attached poles in G: []
-> Checking Foot x[3]: Attached poles in G: ['p[0]']
-> Pathfinding x[1] to x[2]: Found 1 simple paths.
-> Pathfinding x[1] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]
Leg to x[2]: x[1] -> d[1] -> x[2]
Leg to x[3]: x[1] -> d[1] -> x[2] -> d[0] -> x[3]

Evaluating Head Candidate: x[2]
-> Checking Foot x[1]: Attached poles in G: ['p[1]']
-> Checking Foot x[3]: Attached poles in G: ['p[0]']
-> Pathfinding x[2] to x[1]: Found 1 simple paths.
-> Pathfinding x[2] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]
Leg to x[1]: x[2] -> d[1] -> x[1]
Leg to x[3]: x[2] -> d[0] -> x[3]

Evaluating Head Candidate: x[3]
-> Checking Foot x[1]: Attached poles in G: ['p[1]']
-> Checking Foot x[2]: Attached poles in G: []
-> Pathfinding x[3] to x[1]: Found 1 simple paths.
-> Pathfinding x[3] to x[2]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[3]
Leg to x[1]: x[3] -> d[0] -> x[2] -> d[1] -> x[1]
Leg to x[2]: x[3] -> d[0] -> x[2]

$f = 4x^3 - 6x^2 + 3/2$, the number of post critical points is 2

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: $x[1]$

- > Checking Foot $x[2]$: Attached poles in G: []
- > Checking Foot $x[3]$: Attached poles in G: ['p[0]']
- > Pathfinding $x[1]$ to $x[2]$: Found 1 simple paths.
- > Pathfinding $x[1]$ to $x[3]$: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[1]$

Leg to $x[2]$: $x[1] \rightarrow d[0] \rightarrow x[2]$

Leg to $x[3]$: $x[1] \rightarrow d[0] \rightarrow x[2] \rightarrow d[1] \rightarrow x[3]$

Evaluating Head Candidate: $x[2]$

- > Checking Foot $x[1]$: Attached poles in G: ['p[1]']
- > Checking Foot $x[3]$: Attached poles in G: ['p[0]']
- > Pathfinding $x[2]$ to $x[1]$: Found 1 simple paths.
- > Pathfinding $x[2]$ to $x[3]$: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[2]$

Leg to $x[1]$: $x[2] \rightarrow d[0] \rightarrow x[1]$

Leg to $x[3]$: $x[2] \rightarrow d[1] \rightarrow x[3]$

Evaluating Head Candidate: $x[3]$

- > Checking Foot $x[1]$: Attached poles in G: ['p[1]']
- > Checking Foot $x[2]$: Attached poles in G: []
- > Pathfinding $x[3]$ to $x[1]$: Found 1 simple paths.
- > Pathfinding $x[3]$ to $x[2]$: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: $x[3]$

Leg to $x[1]$: $x[3] \rightarrow d[1] \rightarrow x[2] \rightarrow d[0] \rightarrow x[1]$

Leg to $x[2]$: $x[3] \rightarrow d[1] \rightarrow x[2]$

$f = 3x^3 - 9/2x^2 + 1$, the number of post critical points is 2

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: $x[1]$

- > Checking Foot $x[2]$: Attached poles in G: ['p[0]']
- > Checking Foot $x[3]$: Attached poles in G: ['p[0]']
- > Pathfinding $x[1]$ to $x[2]$: Found 1 simple paths.
- > Pathfinding $x[1]$ to $x[3]$: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

Leg to x[2]: x[1] -> d[0] -> x[2]

Leg to x[3]: x[1] -> d[0] -> x[2] -> p[0] -> x[3]

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: ['p[1]']

-> Checking Foot x[3]: Attached poles in G: ['p[0]']

-> Pathfinding x[2] to x[1]: Found 1 simple paths.

-> Pathfinding x[2] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

Leg to x[1]: x[2] -> d[0] -> x[1]

Leg to x[3]: x[2] -> p[0] -> x[3]

Evaluating Head Candidate: x[3]

-> Checking Foot x[1]: Attached poles in G: ['p[1]']

-> Checking Foot x[2]: Attached poles in G: ['p[0]']

-> Pathfinding x[3] to x[1]: Found 1 simple paths.

-> Pathfinding x[3] to x[2]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[3]

Leg to x[1]: x[3] -> p[0] -> x[2] -> d[0] -> x[1]

Leg to x[2]: x[3] -> p[0] -> x[2]

$f = -x^3 + 3/2x^2 - 1$, the number of post critical points is 3

--- Spider Search Results (DIAGNOSTIC MODE) ---

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: ['p[2]', 'p[1]']

[REJECTED] Foot x[2] has > 1 pole. Head x[1] is invalid.

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: []

-> Checking Foot x[3]: Attached poles in G: ['p[0]']

-> Pathfinding x[2] to x[1]: Found 1 simple paths.

-> Pathfinding x[2] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

Leg to x[1]: x[2] -> d[3] -> x[1]

Leg to x[3]: x[2] -> d[3] -> x[1] -> d[2] -> x[3]

Evaluating Head Candidate: x[3]

-> Checking Foot x[1]: Attached poles in G: []
-> Checking Foot x[2]: Attached poles in G: ['p[2]', 'p[1]']
[REJECTED] Foot x[2] has > 1 pole. Head x[3] is invalid.

$f = -1/4 \cdot x^3 + 3/2 \cdot x + 2$, the number of post critical points is 4

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

-> Checking Foot x[2]: Attached poles in G: []
-> Checking Foot x[3]: Attached poles in G: []
-> Pathfinding x[1] to x[2]: Found 1 simple paths.
-> Pathfinding x[1] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

Leg to x[2]: x[1] -> d[2] -> x[2]
Leg to x[3]: x[1] -> d[2] -> x[2] -> d[1] -> x[3]

Evaluating Head Candidate: x[2]

-> Checking Foot x[1]: Attached poles in G: []
-> Checking Foot x[3]: Attached poles in G: []
-> Pathfinding x[2] to x[1]: Found 1 simple paths.
-> Pathfinding x[2] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

Leg to x[1]: x[2] -> d[2] -> x[1]
Leg to x[3]: x[2] -> d[1] -> x[3]

Evaluating Head Candidate: x[3]

-> Checking Foot x[1]: Attached poles in G: []
-> Checking Foot x[2]: Attached poles in G: []
-> Pathfinding x[3] to x[1]: Found 1 simple paths.
-> Pathfinding x[3] to x[2]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[3]

Leg to x[1]: x[3] -> d[1] -> x[2] -> d[2] -> x[1]
Leg to x[2]: x[3] -> d[1] -> x[2]

$f = -1/28 \cdot x^3 - 3/4 \cdot x + 7/2$, the number of post critical points is 4

---- Spider Search Results (DIAGNOSTIC MODE) ----

Evaluating Head Candidate: x[1]

- > Checking Foot x[2]: Attached poles in G: []
- > Checking Foot x[3]: Attached poles in G: []
- > Pathfinding x[1] to x[2]: Found 1 simple paths.
- > Pathfinding x[1] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[1]

- Leg to x[2]: x[1] -> d[8] -> x[3] -> d[7] -> x[2]
- Leg to x[3]: x[1] -> d[8] -> x[3]

Evaluating Head Candidate: x[2]

- > Checking Foot x[1]: Attached poles in G: []
- > Checking Foot x[3]: Attached poles in G: []
- > Pathfinding x[2] to x[1]: Found 1 simple paths.
- > Pathfinding x[2] to x[3]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[2]

- Leg to x[1]: x[2] -> d[7] -> x[3] -> d[8] -> x[1]
- Leg to x[3]: x[2] -> d[7] -> x[3]

Evaluating Head Candidate: x[3]

- > Checking Foot x[1]: Attached poles in G: []
- > Checking Foot x[2]: Attached poles in G: []
- > Pathfinding x[3] to x[1]: Found 1 simple paths.
- > Pathfinding x[3] to x[2]: Found 1 simple paths.

[SUCCESS] VALID SPIDER FOUND! Head: x[3]

- Leg to x[1]: x[3] -> d[8] -> x[1]
- Leg to x[2]: x[3] -> d[7] -> x[2]